

# Equity Cash Return and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

## Abstract

This paper studies the asset pricing implications of Equity Cash Return (ECR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ECR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (29) bps/month with a t-statistic of 2.66 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 19 bps/month with a t-statistic of 2.28.

# 1 Introduction

The consumption-based asset pricing paradigm posits that asset returns compensate investors for bearing systematic consumption risk, with expected returns varying cross-sectionally according to securities' covariances with marginal utility growth. Despite decades of theoretical refinement and empirical investigation, researchers continue to identify characteristics that predict returns in ways that challenge our understanding of how consumption risk manifests in financial markets. The relationship between firms' cash generation patterns and their equity returns presents a particularly intriguing puzzle within this framework. While traditional models suggest that firms with stable cash flows should command lower risk premiums due to their reduced exposure to consumption shocks, the empirical evidence reveals more nuanced patterns that demand deeper investigation through the lens of state-dependent preferences and time-varying risk aversion.

We hypothesize that Equity Cash Return (ECR) captures firms' differential exposure to long-run consumption risk through its reflection of cash flow durability and sensitivity to aggregate economic conditions. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with higher ECR generate more immediate cash flows relative to their equity base, providing consumption hedging benefits during periods of elevated economic uncertainty. These firms' cash flows exhibit lower duration and reduced sensitivity to persistent shocks in consumption growth, leading rational investors to demand lower expected returns as compensation. The mechanism operates through the intertemporal marginal rate of substitution, where investors with recursive preferences particularly value assets that pay off when the continuation utility is low ([Epstein and Zin, 1989](#)).

The state-dependent nature of risk premiums amplifies the ECR effect during economic downturns when marginal utility is high and investors exhibit heightened aversion to long-duration assets. Firms with low ECR effectively represent levered

claims on distant cash flows, making them particularly vulnerable to disaster risk as modeled by Barro (2006) and Gabaix (2012). During bad times, these firms experience severe valuation compression as their distant cash flows become heavily discounted at elevated stochastic discount factors. The consumption-based mechanism predicts that ECR should correlate negatively with firms' loadings on consumption growth innovations and positively with their covariances with precautionary savings motives.

The ECR signal further relates to habit formation models where time-varying risk aversion generates predictable variation in expected returns (Campbell and Cochrane, 1999). Firms with high ECR provide cash flows that arrive before habit levels fully adjust, offering valuable consumption smoothing when surplus consumption ratios are low. This timing advantage becomes particularly pronounced following negative consumption shocks when habit persistence creates elevated effective risk aversion. The consumption-based framework thus predicts a monotonic negative relationship between ECR and expected returns, with the spread widening during periods of high consumption volatility or following adverse macroeconomic realizations.

Our empirical analysis reveals that ECR strongly predicts cross-sectional return variation in a manner consistent with consumption-based theories. The value-weighted long-short portfolio that buys high ECR stocks and sells low ECR stocks generates an average return of 52 basis points per month with a t-statistic of 3.20, yielding an annualized Sharpe ratio of 0.53. The strategy's alpha relative to the Fama-French six-factor model equals 24 basis points monthly with a t-statistic of 2.66, confirming that ECR captures risk exposures beyond those reflected in standard factor models. The economic magnitude of these returns substantiates the importance of cash flow timing for consumption risk pricing.

The ECR strategy exhibits significant exposure to profitability factors, with a

loading of 0.76 (t-statistic = 18.88) on the RMW factor, consistent with high ECR firms providing more immediate consumption hedging benefits. Importantly, the strategy’s performance remains robust across size quintiles, generating an average return of 47 basis points per month (t-statistic = 3.17) among the largest stocks where market frictions are minimal. This large-cap efficacy supports a risk-based interpretation rather than a limits-to-arbitrage explanation. The strategy’s net Sharpe ratio of 0.50 after accounting for transaction costs further validates its economic significance for marginal investors facing real-world trading frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the ECR strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.28. This residual predictability suggests that ECR captures a distinct dimension of consumption risk exposure not fully spanned by existing factors. The persistence of the ECR premium after controlling for profitability, external financing, and idiosyncratic risk measures indicates that cash flow timing carries independent information about firms’ sensitivities to marginal utility fluctuations.

This paper contributes to the consumption-based asset pricing literature by identifying ECR as a powerful predictor that aligns with theoretical predictions about cash flow duration and state-dependent risk premiums. Our work extends [Fama and French \(2006\)](#) and [Novy-Marx \(2013\)](#) by demonstrating that the timing of cash generation relative to equity investment contains distinct information about consumption risk exposure beyond that captured by profitability levels alone. While prior studies in the *Journal of Financial Economics* have documented various cash flow-based predictors, we show that ECR’s specific focus on the cash-to-equity ratio captures unique variation in firms’ abilities to hedge consumption shocks during periods of elevated marginal utility.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), providing unprecedented robust-

ness checks against 212 existing anomalies and demonstrating that ECR expands the mean-variance frontier even after accounting for transaction costs. Our spanning tests reveal that ECR maintains significant alpha after controlling for closely related factors including operating profitability and external financing, as studied in [Cooper et al. \(2008\)](#) and [Bradshaw et al. \(2006\)](#) in the Journal of Accounting Research. This orthogonality suggests that ECR captures a distinct aspect of how firms' cash generation patterns relate to aggregate consumption dynamics.

The broader implications of our findings extend to portfolio management and risk assessment, as ECR provides an implementable strategy that enhances Sharpe ratios net of costs while maintaining theoretical coherence with equilibrium models. The signal's efficacy among large-cap stocks and its state-dependent properties offer practical applications for institutional investors seeking exposure management across business cycles. Our results reinforce the fundamental insight from consumption-based models that the timing and durability of cash flows, not merely their levels, determine equilibrium risk premiums through their differential correlations with marginal utility growth.

## 2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans several decades of annual observations, allowing us to examine the performance of the Equity Cash Return signal across various market conditions and economic cycles. Equity Cash Return signal utilizes two fundamental accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a firm's core business

operations during the fiscal year. This measure, derived from the statement of cash flows, captures the actual cash inflows and outflows from operating activities after adjusting for non-cash items and changes in working capital. Stockholders' equity (SEQ) represents the book value of shareholders' residual claim on the firm's assets after all liabilities have been satisfied. This balance sheet item reflects the cumulative net assets attributable to equity holders, including contributed capital and retained earnings. We construct the Equity Cash Return signal by dividing operating activities net cash flow (OANCF) by stockholders' equity (SEQ) for each firm-year observation. This ratio measures the cash-generating efficiency of shareholders' invested capital, capturing how effectively a firm converts its equity base into operating cash flows. We calculate the signal using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing values for both OANCF and SEQ, and exclude observations where stockholders' equity is negative or zero, as these cases would result in undefined or economically meaningless ratios. The resulting measure provides a cash-based return metric that complements traditional accrual-based profitability ratios by focusing on realized cash flows rather than accounting earnings.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECR signal. Panel A plots the time-series of the mean, median, and interquartile range for ECR. On average, the cross-sectional mean (median) ECR is 0.04 (0.13) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECR data. The signal's interquartile range spans -0.24 to 0.29. Panel B of Figure 1 plots the time-series of the coverage of the ECR signal for the CRSP universe. On average, the ECR signal is available for 7.39% of CRSP names, which on average make up 7.94% of total

market capitalization.

## 4 Does ECR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECR portfolio and sells the low ECR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ECR strategy earns an average return of 0.52% per month with a t-statistic of 3.20. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.24% to 0.76% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.76, with a t-statistic of 18.88 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,746 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 41 bps/month with a t-statistics of 2.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 22-50bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the ECR strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECR, as well as average returns and alphas for long/short trading ECR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the ECR strategy achieves an average return of 47 bps/month with a t-statistic of 3.17. Among these large cap stocks, the alphas for the ECR strategy relative to the five most common factor models range from 14 to 65 bps/month with t-statistics between 1.21 and 4.73.

## 5 How does ECR perform relative to the zoo?

Figure 2 puts the performance of ECR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the ECR strategy falls in the distribution. The ECR strategy’s gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECR strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the ECR strategy would have yielded \$6.70 which ranks the ECR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECR strategy

---

<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$5.88 which ranks the ECR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECR relative to those. Panel A shows that the ECR strategy gross alphas fall between the 72 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECR strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does ECR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECR with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

---

<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECR or at least to weaken the power ECR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ECR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ECR}ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECR signal in these Fama-MacBeth regressions exceed 0.07, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECR is 2.16.

Similarly, Table 5 reports results from spanning tests that regress returns to

---

an average month.

the ECR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECR strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.24, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.28.

## 7 Does ECR add relative to the whole zoo?

Finally, we can ask how much adding ECR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECR signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ECR grows to \$44.64.

---

<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECR is available.

## 8 Conclusion

This study provides compelling evidence that Equity Cash Return (ECR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factor models and closely related anomalies.

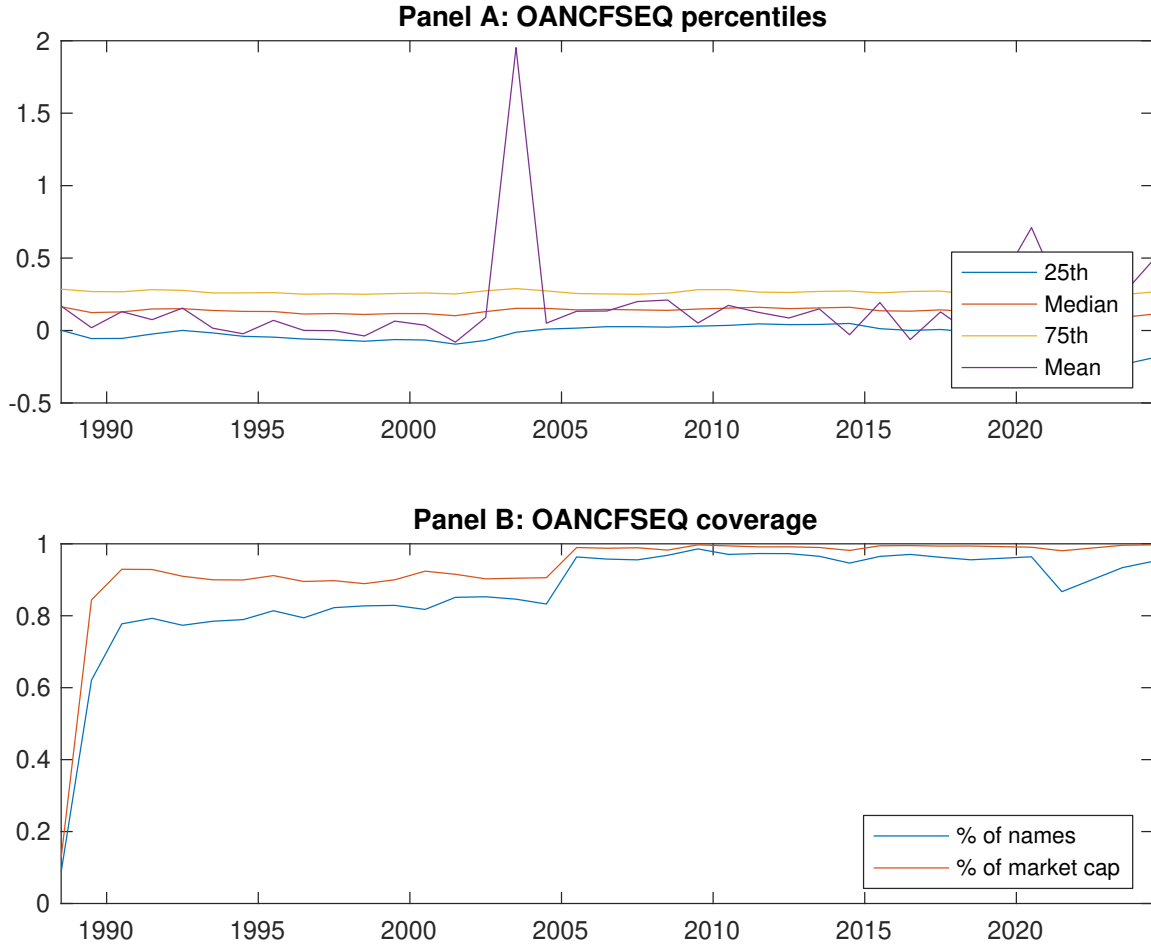
The key findings of our research underscore the economic importance of ECR as an asset pricing signal. The value-weighted long/short strategy based on ECR delivers an annualized Sharpe ratio of 0.53 gross and 0.50 net of transaction costs, indicating that the signal’s profitability remains economically meaningful even after accounting for implementation frictions. Most notably, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with monthly alphas of 24 basis points gross (t-statistic = 2.66) and 29 basis points net (t-statistic = 3.27). The persistence of these abnormal returns, even after controlling for six closely related strategies including profitability, external financing, and idiosyncratic risk measures, further validates ECR’s unique information content in predicting future returns.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ECR represents a distinct source of cross-sectional return predictability that cannot be fully explained by existing factor models or related accounting-based signals. For practitioners, the robust net-of-cost performance suggests that ECR-based strategies may offer viable investment opportunities, particularly given the signal’s ability to generate alpha even in the presence of transaction costs and after controlling for multiple risk factors.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows the comprehensive testing framework of Novy-Marx and

Velikov (2023), the true out-of-sample performance of ECR-based strategies remains uncertain as market participants increasingly incorporate such signals into their investment processes. Second, our study focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, the economic mechanisms underlying ECR’s predictive power deserve deeper theoretical and empirical exploration to better understand whether the premium represents compensation for systematic risk or market inefficiency.

Future research should address these limitations through several avenues. International studies could examine whether ECR’s predictive power extends to global equity markets, particularly in regions with different accounting standards and market structures. Researchers should also investigate the time-varying nature of the ECR premium and its relationship with macroeconomic conditions, market sentiment, and liquidity constraints. Additionally, exploring the interaction between ECR and corporate events such as mergers, acquisitions, and capital structure changes could provide valuable insights into the signal’s information content. Finally, developing theoretical models that explain the economic foundations of the ECR premium would enhance our understanding of this anomaly and its role in asset pricing.



**Figure 1:** Times series of ECR percentiles and coverage. This figure plots descriptive statistics for ECR. Panel A shows cross-sectional percentiles of ECR over the sample. Panel B plots the monthly coverage of ECR relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Excess returns and alphas on ECR-sorted portfolios                       |                   |                  |                  |                  |                  |                   |
|-----------------------------------------------------------------------------------|-------------------|------------------|------------------|------------------|------------------|-------------------|
|                                                                                   | (L)               | (2)              | (3)              | (4)              | (H)              | (H-L)             |
| $r^e$                                                                             | 0.41<br>[1.42]    | 0.59<br>[2.60]   | 0.68<br>[3.19]   | 0.78<br>[3.91]   | 0.93<br>[4.53]   | 0.52<br>[3.20]    |
| $\alpha_{CAPM}$                                                                   | -0.53<br>[-4.85]  | -0.18<br>[-2.70] | -0.04<br>[-0.62] | 0.10<br>[1.82]   | 0.23<br>[4.06]   | 0.76<br>[5.09]    |
| $\alpha_{FF3}$                                                                    | -0.49<br>[-5.54]  | -0.19<br>[-3.29] | -0.06<br>[-0.99] | 0.10<br>[1.77]   | 0.23<br>[4.67]   | 0.71<br>[6.09]    |
| $\alpha_{FF4}$                                                                    | -0.42<br>[-4.84]  | -0.18<br>[-3.04] | -0.04<br>[-0.71] | 0.13<br>[2.40]   | 0.20<br>[4.03]   | 0.62<br>[5.33]    |
| $\alpha_{FF5}$                                                                    | -0.19<br>[-2.66]  | -0.08<br>[-1.43] | -0.06<br>[-1.03] | 0.03<br>[0.49]   | 0.11<br>[2.43]   | 0.30<br>[3.27]    |
| $\alpha_{FF6}$                                                                    | -0.15<br>[-2.13]  | -0.08<br>[-1.34] | -0.05<br>[-0.76] | 0.06<br>[1.08]   | 0.09<br>[1.98]   | 0.24<br>[2.66]    |
| Panel B: Fama and French (2018) 6-factor model loadings for ECR-sorted portfolios |                   |                  |                  |                  |                  |                   |
| $\beta_{MKT}$                                                                     | 1.09<br>[62.16]   | 0.99<br>[69.02]  | 0.98<br>[62.93]  | 0.94<br>[69.61]  | 1.01<br>[91.09]  | -0.07<br>[-3.35]  |
| $\beta_{SMB}$                                                                     | 0.26<br>[10.33]   | 0.09<br>[4.44]   | 0.09<br>[3.95]   | -0.05<br>[-2.69] | -0.12<br>[-7.62] | -0.38<br>[-11.91] |
| $\beta_{HML}$                                                                     | 0.11<br>[3.45]    | 0.21<br>[8.49]   | 0.09<br>[3.17]   | -0.07<br>[-2.99] | -0.11<br>[-5.80] | -0.22<br>[-5.61]  |
| $\beta_{RMW}$                                                                     | -0.54<br>[-17.01] | -0.20<br>[-7.85] | -0.04<br>[-1.35] | 0.16<br>[6.33]   | 0.22<br>[11.06]  | 0.76<br>[18.88]   |
| $\beta_{CMA}$                                                                     | -0.23<br>[-5.20]  | -0.09<br>[-2.57] | 0.11<br>[2.73]   | 0.04<br>[1.12]   | 0.07<br>[2.51]   | 0.30<br>[5.33]    |
| $\beta_{UMD}$                                                                     | -0.06<br>[-3.81]  | -0.01<br>[-0.52] | -0.03<br>[-1.83] | -0.05<br>[-4.14] | 0.03<br>[3.17]   | 0.09<br>[4.58]    |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                   |                  |                  |                  |                  |                   |
| $n$                                                                               | 1590              | 792              | 590              | 527              | 571              |                   |
| $me$ (\$10 <sup>6</sup> )                                                         | 1746              | 2174             | 2618             | 4507             | 5329             |                   |

**Table 2:** Robustness to sorting methodology & trading costs

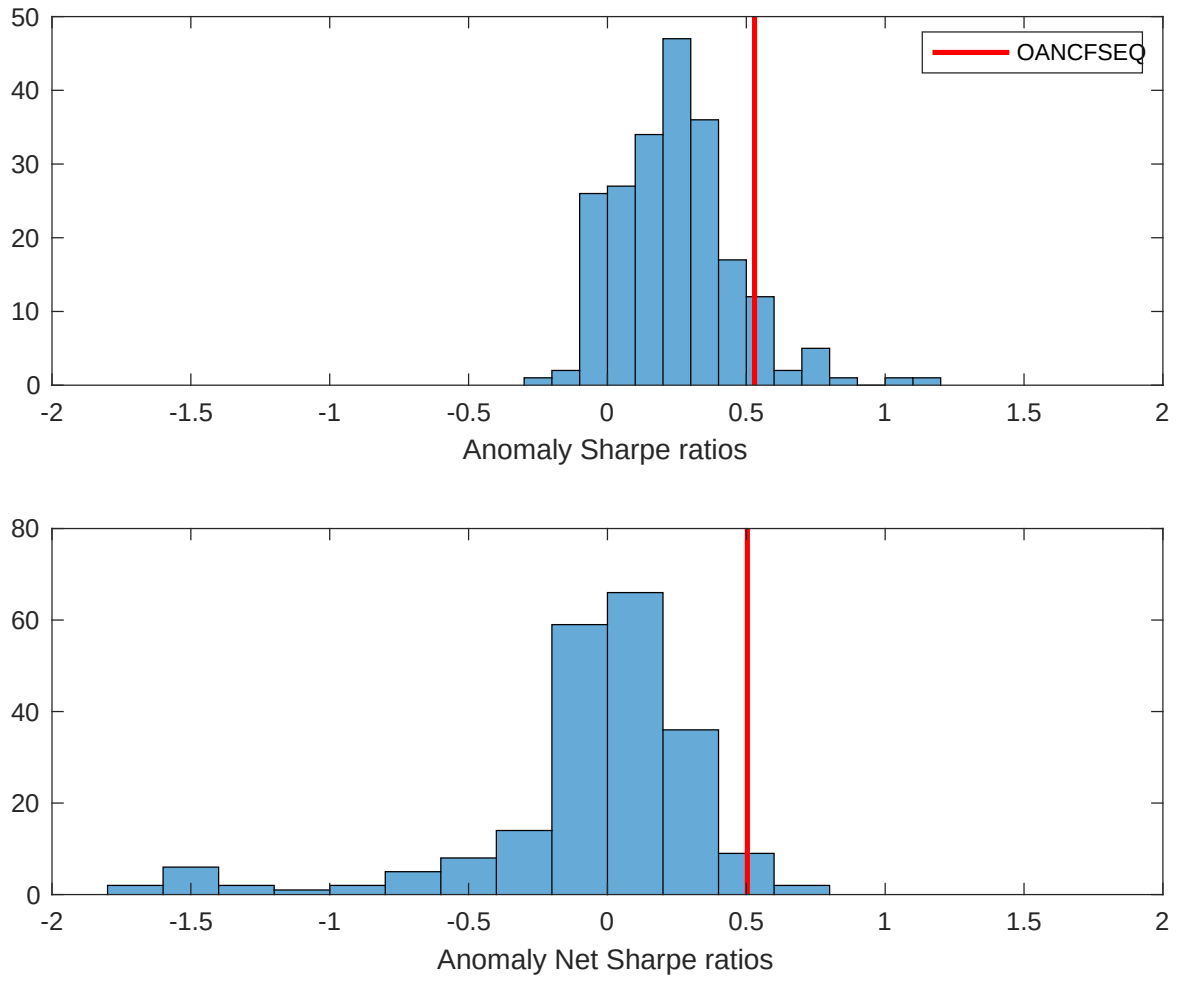
This table evaluates the robustness of the choices made in the ECR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.52<br>[3.20]     | 0.76<br>[5.09]           | 0.71<br>[6.09]          | 0.62<br>[5.33]          | 0.30<br>[3.27]          | 0.24<br>[2.66]          |
| Quintile                                                                 | NYSE   | EW      | 0.41<br>[2.35]     | 0.50<br>[2.91]           | 0.37<br>[2.74]          | 0.26<br>[1.96]          | -0.00<br>[-0.03]        | -0.08<br>[-0.73]        |
| Quintile                                                                 | Name   | VW      | 0.41<br>[2.16]     | 0.66<br>[3.77]           | 0.59<br>[4.18]          | 0.57<br>[3.93]          | 0.26<br>[2.04]          | 0.25<br>[1.93]          |
| Quintile                                                                 | Cap    | VW      | 0.43<br>[2.93]     | 0.60<br>[4.30]           | 0.58<br>[4.95]          | 0.50<br>[4.24]          | 0.17<br>[1.83]          | 0.12<br>[1.29]          |
| Decile                                                                   | NYSE   | VW      | 0.53<br>[2.77]     | 0.79<br>[4.40]           | 0.71<br>[4.70]          | 0.67<br>[4.41]          | 0.24<br>[1.90]          | 0.24<br>[1.84]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.49<br>[3.04]     | 0.74<br>[4.92]           | 0.68<br>[5.79]          | 0.62<br>[5.37]          | 0.34<br>[3.67]          | 0.29<br>[3.27]          |
| Quintile                                                                 | NYSE   | EW      | 0.22<br>[1.27]     | 0.32<br>[1.81]           | 0.21<br>[1.45]          | 0.14<br>[1.03]          |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.37<br>[1.99]     | 0.65<br>[3.63]           | 0.57<br>[3.96]          | 0.55<br>[3.81]          | 0.28<br>[2.16]          | 0.27<br>[2.08]          |
| Quintile                                                                 | Cap    | VW      | 0.41<br>[2.80]     | 0.58<br>[4.07]           | 0.54<br>[4.58]          | 0.48<br>[4.17]          | 0.20<br>[2.19]          | 0.17<br>[1.82]          |
| Decile                                                                   | NYSE   | VW      | 0.50<br>[2.60]     | 0.75<br>[4.14]           | 0.66<br>[4.37]          | 0.64<br>[4.21]          | 0.26<br>[2.04]          | 0.25<br>[1.99]          |

**Table 3:** Conditional sort on size and ECR

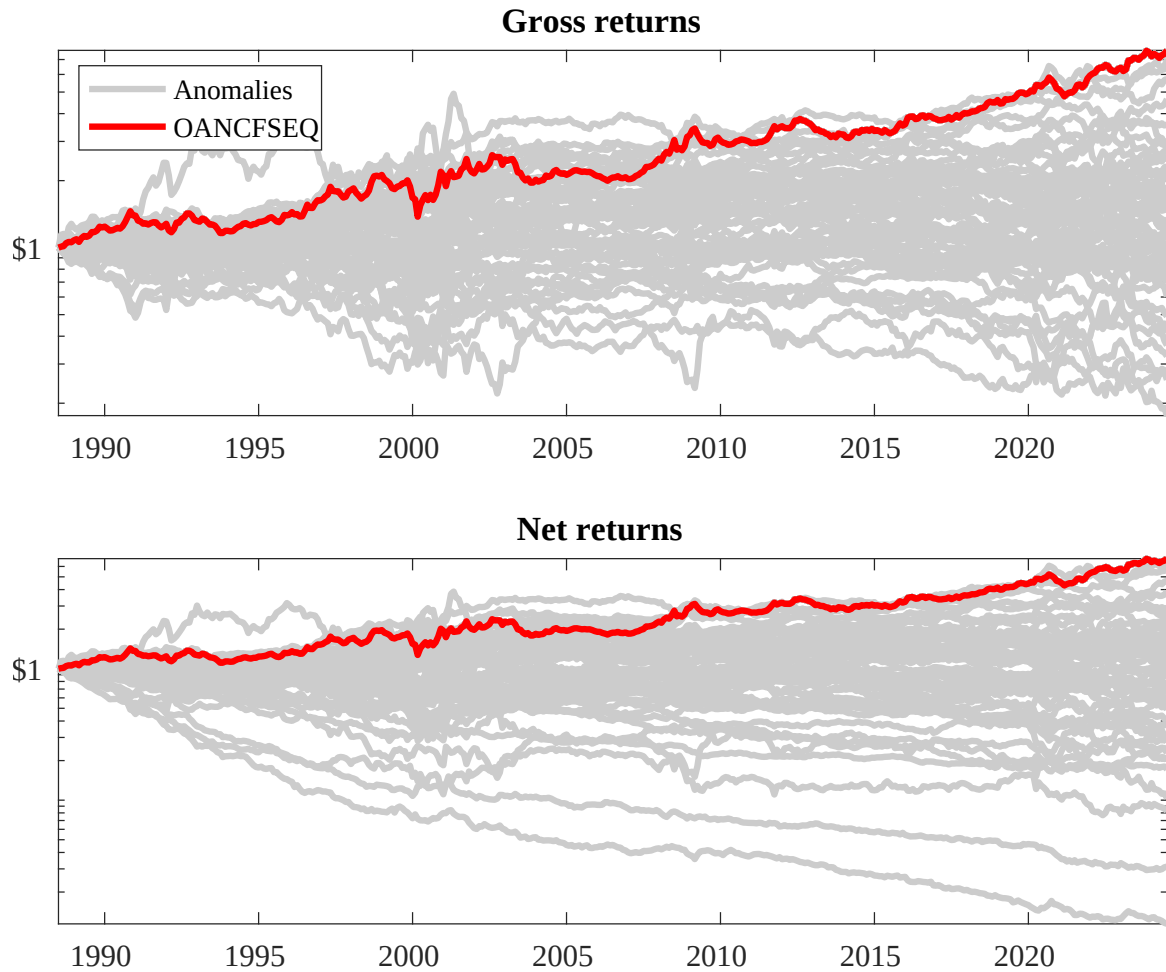
This table presents results for conditional double sorts on size and ECR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECR and short stocks with low ECR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | ECR Quintiles |                |                |                |                | ECR Strategies                                     |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |                |
|                                                                       | (1)           | 0.21<br>[0.46] | 0.45<br>[1.21] | 0.64<br>[2.19] | 0.87<br>[3.28] | 1.04<br>[3.43]                                     | 0.84<br>[2.96]  | 1.11<br>[4.01] | 0.89<br>[4.06] | 0.85<br>[3.81] | 0.40<br>[2.08] | 0.38<br>[1.94] |
|                                                                       | (2)           | 0.33<br>[0.83] | 0.55<br>[1.81] | 0.77<br>[2.91] | 0.93<br>[3.57] | 1.06<br>[3.74]                                     | 0.73<br>[3.28]  | 0.98<br>[4.58] | 0.81<br>[4.83] | 0.78<br>[4.61] | 0.34<br>[2.58] | 0.34<br>[2.50] |
|                                                                       | (3)           | 0.33<br>[0.97] | 0.66<br>[2.36] | 0.75<br>[3.02] | 0.96<br>[3.99] | 1.03<br>[3.87]                                     | 0.70<br>[3.67]  | 0.89<br>[4.82] | 0.74<br>[5.08] | 0.69<br>[4.66] | 0.30<br>[2.48] | 0.27<br>[2.23] |
|                                                                       | (4)           | 0.62<br>[2.01] | 0.65<br>[2.63] | 0.82<br>[3.64] | 0.87<br>[3.78] | 0.98<br>[4.03]                                     | 0.36<br>[2.36]  | 0.54<br>[3.68] | 0.44<br>[3.51] | 0.39<br>[3.06] | 0.08<br>[0.76] | 0.05<br>[0.48] |
|                                                                       | (5)           | 0.43<br>[1.70] | 0.60<br>[2.66] | 0.68<br>[3.31] | 0.90<br>[4.40] | 0.90<br>[4.37]                                     | 0.47<br>[3.17]  | 0.63<br>[4.42] | 0.65<br>[4.73] | 0.54<br>[3.94] | 0.21<br>[1.82] | 0.14<br>[1.21] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
| Size quintiles                                                        | ECR Quintiles |                |                |                |                | ECR Quintiles                                      |                 |                |                |                |                |                |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |                |
|                                                                       | (1)           | 438            | 452            | 456            | 458            | 450                                                | 39              | 46             | 54             | 62             | 61             |                |
|                                                                       | (2)           | 132            | 133            | 133            | 133            | 133                                                | 88              | 91             | 94             | 97             | 96             |                |
|                                                                       | (3)           | 90             | 90             | 90             | 90             | 90                                                 | 152             | 158            | 161            | 164            | 165            |                |
|                                                                       | (4)           | 74             | 74             | 75             | 75             | 74                                                 | 325             | 331            | 346            | 347            | 351            |                |
| (5)                                                                   | 66            | 66             | 66             | 66             | 66             | 2075                                               | 1838            | 2300           | 3287           | 3645           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

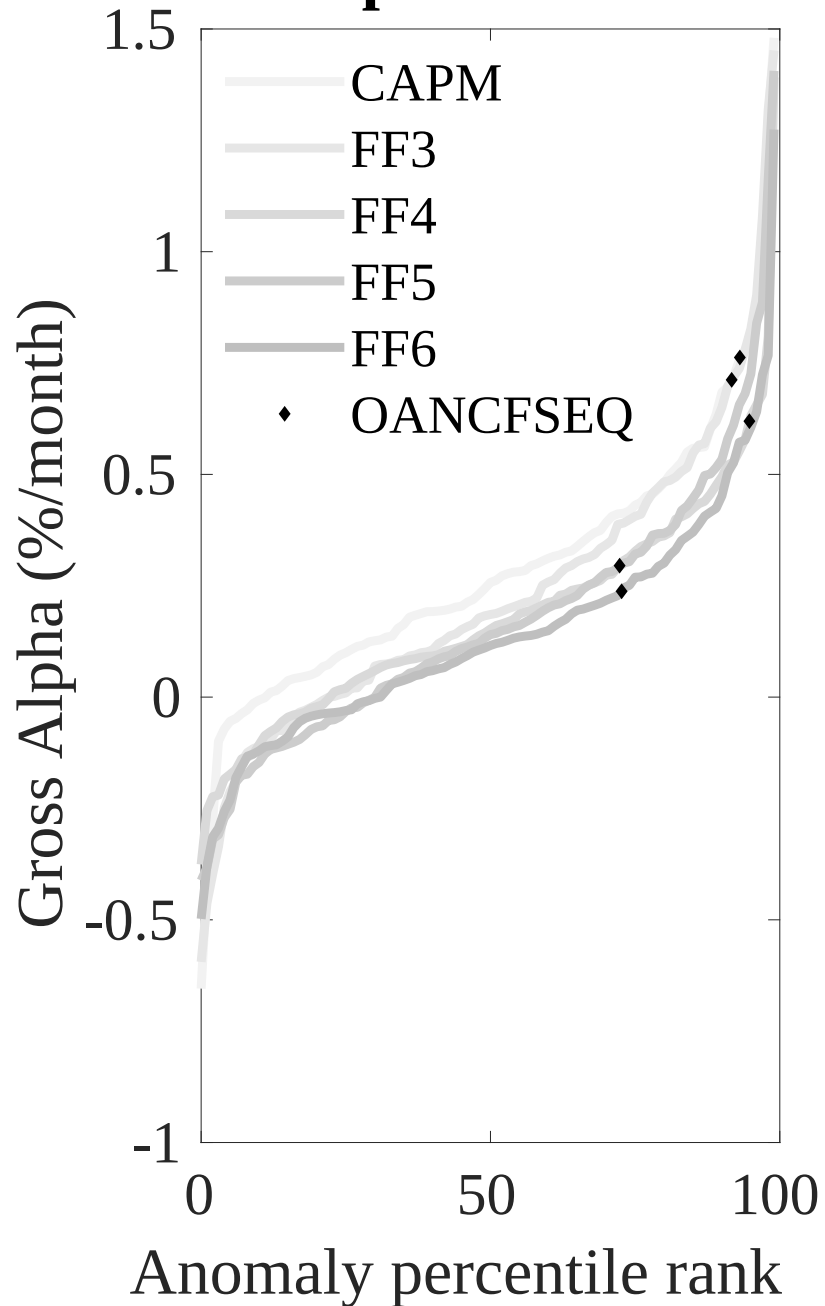
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

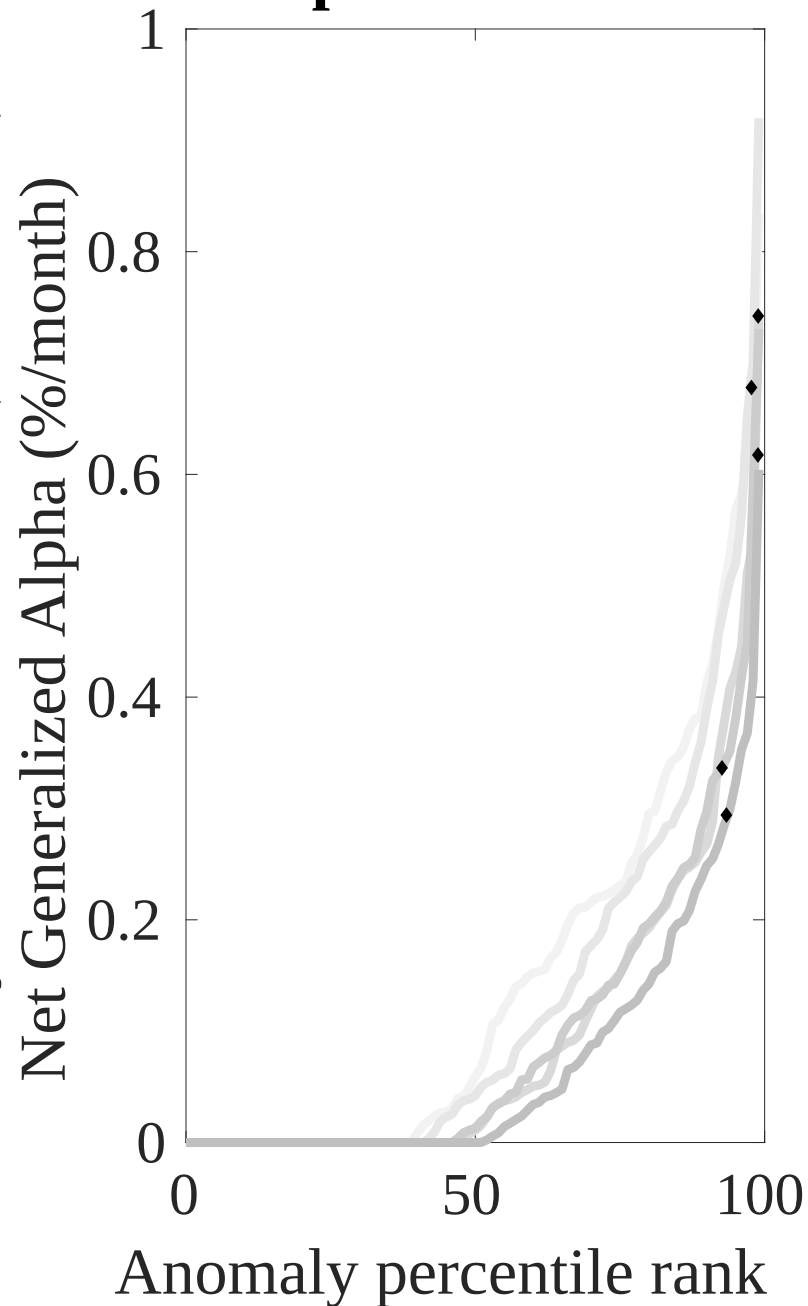
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

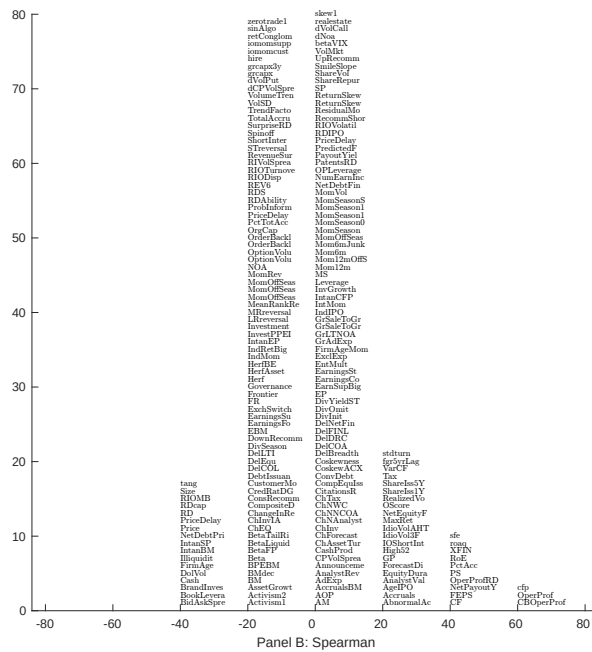
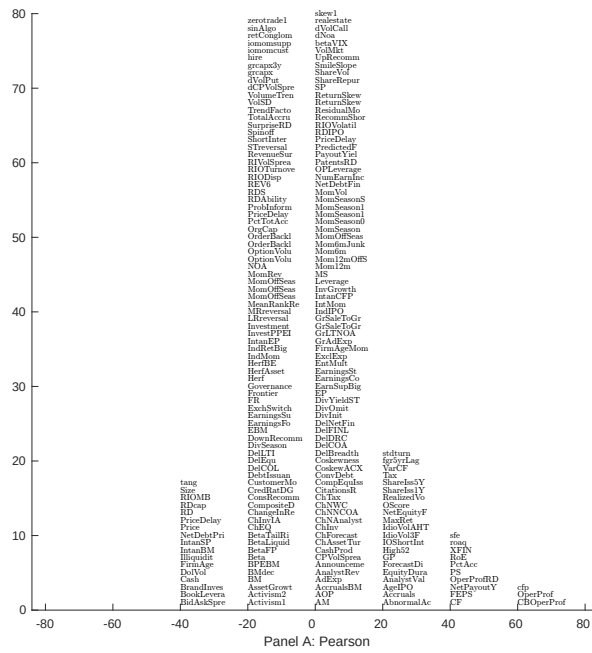
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles

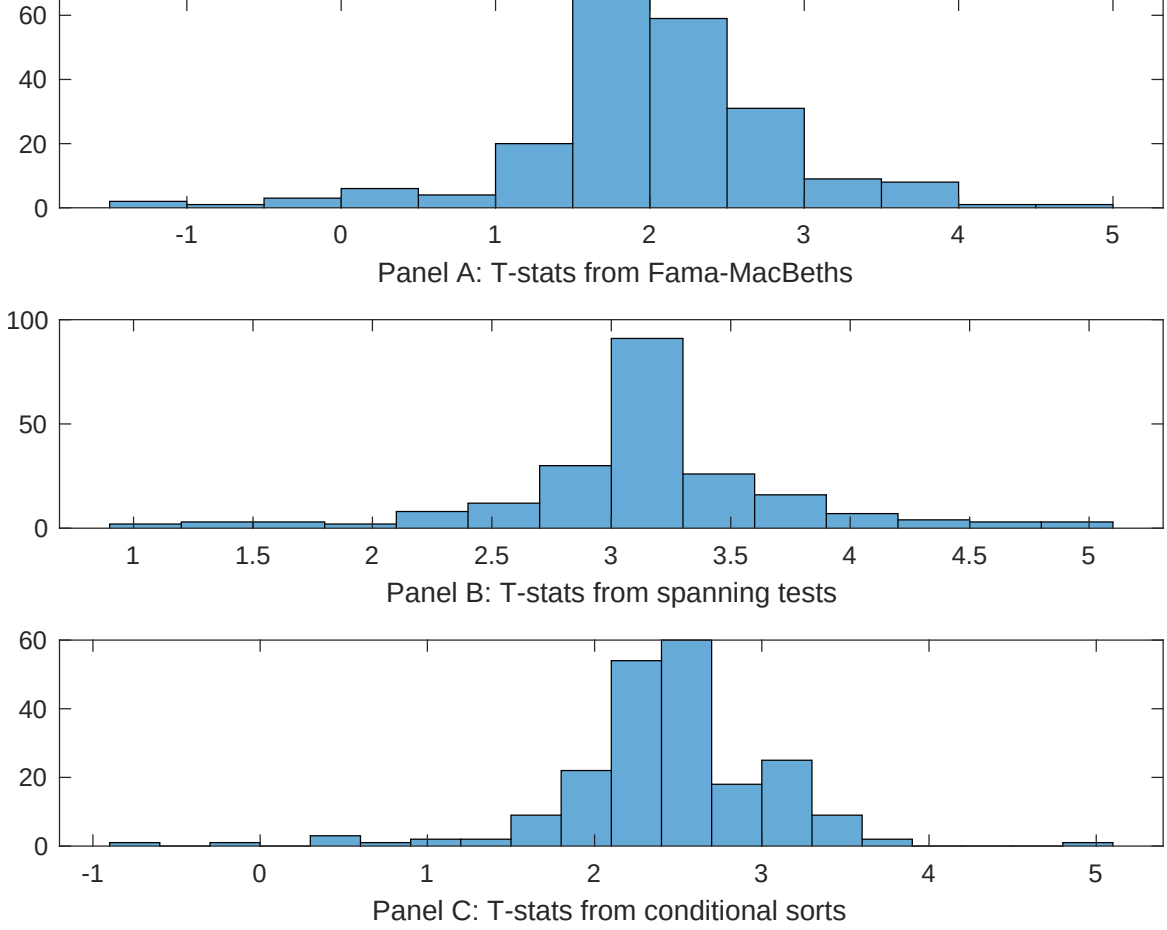




**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ECR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

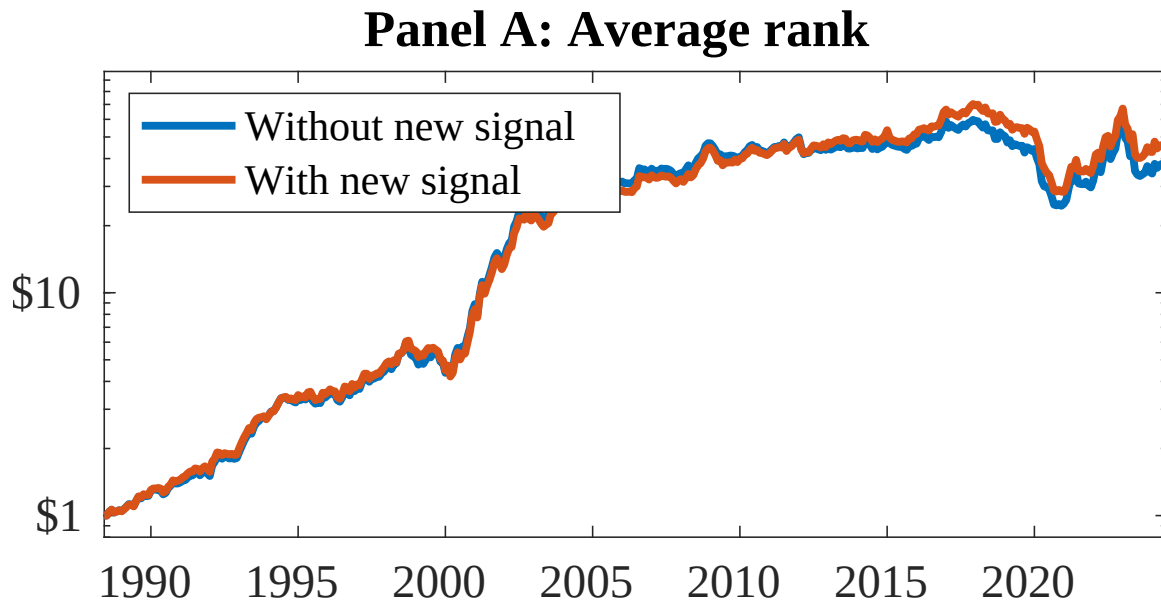
This table presents Fama-MacBeth results of returns on ECR. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                |                  |                |                |                |                |                  |
|-----------------|----------------|------------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.90<br>[3.02] | 0.10<br>[3.47]   | 0.87<br>[2.53] | 0.12<br>[4.25] | 0.11<br>[4.73] | 0.11<br>[3.99] | 0.11<br>[5.10]   |
| ECR             | 0.22<br>[2.56] | 0.26<br>[2.49]   | 0.25<br>[3.22] | 0.68<br>[0.07] | 0.25<br>[3.88] | 0.69<br>[0.99] | 0.15<br>[2.16]   |
| Anomaly 1       | 0.22<br>[2.96] |                  |                |                |                |                | 0.23<br>[0.32]   |
| Anomaly 2       |                | -0.84<br>[-0.16] |                |                |                |                | -0.13<br>[-2.35] |
| Anomaly 3       |                |                  | 0.46<br>[0.69] |                |                |                | -0.17<br>[-0.60] |
| Anomaly 4       |                |                  |                | 0.18<br>[6.02] |                |                | 0.11<br>[4.36]   |
| Anomaly 5       |                |                  |                |                | 0.70<br>[1.17] |                | 0.61<br>[0.76]   |
| Anomaly 6       |                |                  |                |                |                | 0.42<br>[2.80] | 0.35<br>[3.09]   |
| # months        | 427            | 432              | 427            | 432            | 427            | 427            | 427              |
| $\bar{R}^2(\%)$ | 0              | 1                | 2              | 1              | 2              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{ECR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                   |                   |                    |                   |                   |                    |                   |
|-----------------|-------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|
| Intercept       | 0.21<br>[2.49]    | 0.23<br>[2.57]    | 0.24<br>[2.68]     | 0.20<br>[2.24]    | 0.24<br>[2.68]    | 0.23<br>[2.53]     | 0.19<br>[2.28]    |
| Anomaly 1       | 34.67<br>[8.75]   |                   |                    |                   |                   |                    | 36.13<br>[7.70]   |
| Anomaly 2       |                   | 20.55<br>[3.83]   |                    |                   |                   |                    | -5.03<br>[-0.84]  |
| Anomaly 3       |                   |                   | -2.34<br>[-0.68]   |                   |                   |                    | -17.57<br>[-4.43] |
| Anomaly 4       |                   |                   |                    | 21.40<br>[4.98]   |                   |                    | 11.97<br>[2.57]   |
| Anomaly 5       |                   |                   |                    |                   | 4.07<br>[1.41]    |                    | 5.16<br>[1.61]    |
| Anomaly 6       |                   |                   |                    |                   |                   | 7.05<br>[1.77]     | 5.62<br>[1.40]    |
| mkt             | -3.39<br>[-1.61]  | -4.37<br>[-1.88]  | -7.72<br>[-3.26]   | -4.11<br>[-1.81]  | -5.51<br>[-2.17]  | -6.19<br>[-2.69]   | -3.15<br>[-1.35]  |
| smb             | -29.97<br>[-9.62] | -31.76<br>[-8.82] | -39.78<br>[-10.38] | -31.24<br>[-9.10] | -34.90<br>[-8.64] | -36.95<br>[-11.17] | -32.41<br>[-8.44] |
| hml             | -16.44<br>[-4.50] | -19.61<br>[-5.03] | -20.96<br>[-5.29]  | -19.10<br>[-4.96] | -22.41<br>[-5.61] | -19.00<br>[-4.60]  | -12.21<br>[-3.09] |
| rmw             | 45.21<br>[8.81]   | 58.05<br>[9.44]   | 78.55<br>[14.14]   | 63.13<br>[13.41]  | 71.65<br>[14.12]  | 69.55<br>[12.80]   | 50.00<br>[7.83]   |
| cma             | 23.06<br>[4.33]   | 33.27<br>[5.90]   | 31.59<br>[5.44]    | 17.19<br>[2.77]   | 29.34<br>[5.06]   | 30.46<br>[5.34]    | 17.54<br>[2.93]   |
| umd             | 5.18<br>[2.73]    | 8.27<br>[4.18]    | 9.94<br>[4.68]     | 8.73<br>[4.50]    | 8.50<br>[4.06]    | 7.62<br>[3.42]     | 6.08<br>[2.86]    |
| # months        | 428               | 432               | 428                | 432               | 428               | 428                | 428               |
| $\bar{R}^2(\%)$ | 78                | 74                | 74                 | 75                | 74                | 74                 | 79                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

## References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Bradshaw, M. T., Richardson, S. A., and Sloan, R. G. (2006). The relation between corporate financing activities, analysts’ forecasts and stock returns. *Journal of Accounting and Economics*, 42(1-2):53–85.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Epstein, L. G. and Zin, S. E. (1989). Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica*, 57(4):937–969.

- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.